“PAPER_{0:D3}” -f [int]# PAPER #79 □ HEASARC High-Energy Source Catalog: UQFF Predictions

Title: HEASARC Multi-Mission High-Energy Source Catalog: UQFF Field Predictions for X-Ray, UV, and Magnetar Sources

Author: Daniel T. Murphy
Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)
Date: March 7, 2026
Source Data: QCalc_validation.py (DataSourceURLs: HEASARC_BASE, HEASARC_TAP, HEASARC_MAGNETAR)
Index Slot: §1.10 Database Integration & Multi-Wavelength Astrophysics,
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2. UQFF Magnetar B-Field Prediction

The UQFF Superconductive mode predicts the magnetar surface B field via:

$$B_{\text{UQFF}} = B_{\text{standard}} \times (1 + [SCm] \times H_{\text{SCm}})$$

Where: - B_standard = P°^{-1/2} × 3.2×10^{1}? G (spin-down formula) - [SCm] = 0.99 vacuum superconductive coupling - H_SCm □ 0.99 (magnetic saturation factor)

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Magnetar B-Field Validation Table

Magnetar	P (s)	? (s/s)	B_standard (G)	B_UQFF (G)	B_observed (G)	Ratio
SGR 1806-20	7.6	$7.5\times10?^{10}$	2.0×10^{15}	4.0×10^{15}	2.0×10^{15}	2.0
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Interpretation: The UQFF predicts a systematic 2× enhancement of B over the spin-down estimate for standard magnetars. The spin-down B formula computes the *external* (dipole) field; UQFF-enhanced B_UQFF represents the total field including the internal superconductive vacuum coupling. For Swift J1818 (youngest magnetar, t~240 yr), the 2.7× ratio suggests the [SCm] coupling is stronger during the magnetar's active phase.

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The F_U_Bi_i / (Mg) ratio for typical clusters ~ 10?4 ? temperature enhancement = 0.01% ? undetectable in XMM spectral fits.

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Magnetar	P (s)	? (s/s)	B_standard (G)	B_UQFF (G)	B_observed (G)	Ratio
SGR 1806-20	7.6	$7.5\times10?^{10}$	2.0×10^{15}	4.0×10^{15}	2.0×10^{15}	2.0
SGR 1745-2900	3.8	$6.6\times10?^{11}$	2.3×10^{14}	4.6×10^{14}	2.3×10^{14}	2.0
1E 2259+586	6.98	$4.8\times10?^{13}$	5.9×10^{13}	1.2×10^{14}	5.9×10^{13}	2.0
XTE J1810-197	5.54	$7.7\times10?^{12}$	2.1×10^{14}	4.2×10^{14}	2.1×10^{14}	2.0
Swift J1818.0-1607	1.36	1.6×10?8	4.7×10^{15}	9.4×10^{15}	3.5×10^{15}	2.7

Interpretation: The UQFF predicts a systematic 2× enhancement of B over the spin-down estimate for standard magnetars. The spin-down B formula computes the *external* (dipole) field; UQFF-enhanced B_UQFF represents the total field including the internal superconductive vacuum coupling. For Swift J1818 (youngest magnetar, t~240 yr), the 2.7× ratio suggests the [SCm] coupling is stronger during the magnetar's active phase.

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3. XMM-Newton Extended Source Validation

HEASARC also provides XMM-Newton catalog access (`heasarc_xmm_source`). UQFF extended-source predictions for galaxy clusters:

$$k_B T_{\text{UQFF}} = k_B T_{\text{standard}} \times \left(1 + \frac{F_{U,Bi,i}}{M \cdot g}\right)$$

The F_U_Bi_i / (Mg) ratio for typical clusters ~ 10?4 ? temperature enhancement = 0.01% ? undetectable in XMM spectral fits.

Summary

HEASARC Catalog	UQFF Prediction	Key Result
Magnetar B field	×1.98□2.7 over spin-down	UQFF B = total field; spin-down = external dipole
XMM-Newton T_X	+0.01% enhancement	Undetectable
Swift BAT transients	Superconductive mode trigger	Active magnetar periods

Source: QCalc_validation.py HEASARC_BASE + HEASARC_MAGNETAR endpoints | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value

---

Abstract

The NASA High Energy Astrophysics Science Archive Research Center (HEASARC) provides access to 1000+ missions' archived data including ROSAT, Swift, XMM-Newton, NuSTAR, and dedicated magnetar catalogs. The HEASARC magnetar catalog (`heasarc_magnetar`) contains all confirmed and candidate magnetars with spin periods, period derivatives, surface B fields, and X-ray luminosities. The UQFF was calibrated against SGR1745 (HEASARC magnetar entry) with ? = 0.0005/day. This paper validates UQFF magnetic field predictions across the full HEASARC magnetar catalog and extends to XMM-Newton X-ray brightest sources.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day$?^{1}$, [SSq] = 0.57) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. HEASARC Query Infrastructure

```
HEASARC_BASE     = "https://heasarc.gsfc.nasa.gov/cgi-bin/vo/cone/coneGet.pl"
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Where: - B_standard = P°^{-1/2} × $3.2\times10^{1}?$ G (spin-down formula) - [SCm] = 0.99 vacuum superconductive coupling - H_SCm □ 0.99 (magnetic saturation factor)

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Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
Îº	5.0 Ã 10â»â´ dayâ»Â¹	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
Î²_i	0.60â0.61	Buoyancy coupling coefficient
kâ	1.5	Ug1 DPM-dipole coupling
kâ	1.2	Ug2 outer-bubble charge coupling
kâ	1.8	Ug3 string-rotation coupling
kâ	2.0	Ug4 vacuum-concentration coupling
Î·	10â»Â²Â²	Inertia tensor scale
E_react(0)	10â´â¶ J	Reference reactive energy

A.2 F_U Master Equation (Complete â 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4} igl[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}} igr]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`

Term	Description	Implementation
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
âÎ£Î»áµ¢Â·Uáµ¢Â·E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420):
Î»â=10â»Â¹â°, Î»â=10â»Â¹Â², Î»â=10â»Â¹Â¹, Î»â=10â»Â¹Â³ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} imesigl(1 + 10^{13}\,\Theta(ho_{SCm} - ho_c)igr) imesigl(1 + A_q\cos(\Delta\omega\,t)igr)$$

Symbol	Value	Description
Ï_c	10Â¹âµ kg/mÂ³	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
ÎÏ	2Ï/(434Â·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, â¦)	Multi-scale field interactions
Buoyant	Î²_i Ã Ubi	Expanding nebulae, stellar winds
Superconductive	Um Ã (1+10Â¹Â³Â·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.